

Stewart C. Jamieson

PHD STUDENT · AUTONOMOUS ROBOTS & MACHINE LEARNING

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A roboticist working to develop autonomous systems that co-operate with humans in uncertain, unstructured, and unknown remote environments and accomplish their objectives safely and efficiently.

Research Interests & Skills

Artificial Intelligence	Online & Active Learning, Unsupervised Learning, Robust Vision, Spatiotemporal Topic Modelling, AI Ethics
Robotics	Goal-Directed Exploration, Human-Robot Interaction, Uncertainty Quantification, Informative Path Planning
Programming	C++20, Python, PyTorch, TensorFlow, ROS, OpenCV, Pandas, MATLAB, Bash, Java, Android

Education

Massachusetts Institute of Technology & Woods Hole Oceanographic Institution

Cambridge, MA, USA

PH.D. IN APPLIED OCEAN SCIENCE AND ENGINEERING — MARINE ROBOTICS (5.0/5.0 CGPA)

June 2020 - Present

- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)
- Developing multi-robot systems that collaboratively explore remote, unfamiliar environments with limited human guidance

S.M. IN AERONAUTICS AND ASTRONAUTICS — AUTONOMOUS SYSTEMS (5.0/5.0 CGPA)

June 2018 - May 2020

- Thesis: *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*
- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)
- Relevant Coursework: Cognitive Robotics, Visual Navigation for Autonomous Vehicles, Bayesian Modelling and Inference

University of Toronto

Toronto, ON, Canada

B.A.SC. IN ENGINEERING SCIENCE WITH HONOURS — ROBOTICS MAJOR (3.83/4.0 CGPA)

Sept. 2013 - Apr. 2018

- Thesis: *Deep Learning for Robust Vision in Realtime Autonomous Driving*, supervised by Prof. Angela Schoellig
- Applied state-of-the-art uncertainty quantification techniques to achieve safer autonomous driving performance

Work & Research Experience

WHOI's Autonomous Robotics and Perception Laboratory (WARPLab)

Woods Hole, MA, USA

GRADUATE RESEARCH ASSISTANT

June 2018 - Present

- Developing autonomous exploration algorithms for multi-robot teams exploring the deep ocean and coral reefs
- Publications focus on enabling robot co-operation with humans over slow, bandwidth-limited communication channels
- Assisting with the deployment of these novel algorithms into WHOI's world-class deep sea exploration vehicles

aUToronto Self-Driving Car Team (SAE/GM AutoDrive Challenge)

Toronto, ON, Canada

SOFTWARE TEAM LEAD

June 2017 - June 2018

- Led a subteam of 12 graduate and undergraduate students working to develop an autonomous Chevrolet Bolt
- My team created the overall system software architecture, sensor drivers, vehicle control interface, and software services
- At the end of my term as lead, aUToronto won 1st place in Year One of the SAE/GM AutoDrive Challenge

Zebra Technologies Inc.

Mississauga, ON, Canada

SOFTWARE ENGINEERING INTERN, ENGINEERING PRODUCT INNOVATION TEAM

May 2016 - Aug. 2017

- Co-designed and co-developed the software of the first Zebra SmartSight™ prototype, a robot designed for retail operations
- Helped to research and present business applications for robotics, machine learning, and neural networks
- 16 months of C++14 development experience including networking, databases, and high performance computing

Wattpad Inc.

Toronto, ON, Canada

ANDROID SOFTWARE DEVELOPER, READER ACQUISITION TEAM

May 2015 - Sept. 2015

- Wattpad is a worldwide storytelling platform with a community of over 80 million users
- Implemented features designed to attract new users; also implemented A/B tests to validate each features' success

QA SOFTWARE DEVELOPER, ANDROID CORE TEAM

May 2014 - Sept. 2014

- Searched for, reported, and fixed software bugs in the Android mobile application with over 15 million users
- Designed and implemented a virtual doorman to greet company visitors and notify staff of their arrival

Publications

PEER-REVIEWED CONFERENCE PAPERS

Jamieson, S., Fathian, K., Khosoussi, K., How, J. P., Girdhar, Y. (2021). *Multi-Robot Distributed Semantic Mapping in Unfamiliar Environments through Online Matching of Learned Representations*. In 2021 IEEE International Conference on Robotics and Automation (ICRA). Xi'an, China.

Jamieson, S., How, J. P., Girdhar, Y. (2020). *Active Reward Learning for Co-Robotic Vision Based Exploration in Bandwidth Limited Environments*. In 2020 IEEE International Conference on Robotics and Automation (ICRA). Paris, France. **Won Best Paper Award in Service Robotics**.

Girdhar, Y., Cai, L., **Jamieson, S.**, McGuire, N., Flaspohler, G., Suman, S., & Claus, B. (2019). *Enabling Co-Robotic Scientific Exploration of Unknown Environments over a Low Bandwidth Communication Channel*. In 2019 IEEE International Conference on Robotics and Automation (ICRA). Montréal, Canada.

WORKSHOP PAPERS

Jamieson, S., Todd, J. E., How, J. P., Girdhar, Y. (2021). *Communicating Efficiently to Enable Human-Multi-Robot Collaboration in Space Exploration*. In "SpaceCHI: Human-Computer Interaction for Space Exploration" Workshop at CHI 2021. Yokohama, Japan.

Beaulieu, S., Alexander, H., **Jamieson, S.**, Longworth, B., McLean, C., Soenen, K., York, A., Krinos, A., Cai, L., Govostes, R. and Hernandez, C. (2020). *Building a data science curriculum and community for ocean scientists, engineers, and students using The Carpentries model*. In AGU Fall Meeting 2020.

Jamieson, S. (2019). *The Pervasiveness of Deep Learning in Robotics Research Does Not Impede Scientific Insights into Robotics Problems*. In "Debates on the Future of Robotics Research" Workshop at ICRA 2019. Montréal, Canada.

THESES

Jamieson, S. (2020). *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*. Master's Thesis, Massachusetts Institute of Technology & Woods Hole Oceanographic Institution.

Jamieson, S. (2018). *Deep Learning for Robust Vision in Realtime Autonomous Driving*. B.A.Sc. Thesis, University of Toronto.

Honors & Awards

INTERNATIONAL

2020 **Best Paper Award in Service Robotics (out of 1483 Accepted Papers)**, ICRA 2020

[Paris, France](#)

2018 **1st Place Team**, SAE/GM AutoDrive Challenge

[Yuma, AZ, USA](#)

DOMESTIC

2018 **Dean's Honour List**, University of Toronto

[Toronto, ON, Canada](#)

2014-16 **Dean's Honour List (x3)**, University of Toronto

[Toronto, ON, Canada](#)

2013 **Governor General's Bronze Medal for Academic Excellence**, High School Graduation

[Burlington, ON, Canada](#)

2013 **Regional Champion**, ECOO Programming Competition

[Halton, ON, Canada](#)

2010-13 **School Champion (x4)**, Waterloo CEMC Math Contest

[Burlington, ON, Canada](#)

Presentations

Videos and other materials used in some of the following presentations are available at www.stewartjamieson.com

Communicating Efficiently to Enable Human-Multi-Robot Collaboration in Space Exploration

[Online](#)

POSTER PRESENTATION, SPACECHI WORKSHOP @ CHI 2021

[May 2021](#)

Human-Robot Cooperation for Exploring Bandwidth-Limited Environments

[Woods Hole, MA, USA](#)

INVITED TALK, WHOI AOE DEPARTMENT SEMINAR SERIES

[July 2020](#)

Active Reward Learning for Co-Robotic Exploration in Bandwidth-Limited Environments

[Cambridge, MA, USA](#)

INVITED PRESENTER & PANELIST, ICRAxMIT

[June 2020](#)

Deep Learning Does Not Impede Scientific Insights into Robotics Problems

INVITED LIGHTNING TALK, DEBATES ON THE FUTURE OF ROBOTICS RESEARCH, ICRA 2019

16.412 Lecture: Multi-Robot Adaptive Sampling

CO-LECTURER, MASSACHUSETTS INSTITUTE OF TECHNOLOGY

An Introduction to Neural Networks and Machine Learning

LEAD PRESENTER FOR ZEBRA TECHNOLOGIES INC. "LUNCH & LEARN" (200+ ATTENDEES)

Should Robots Have Rights?

CO-PRESENTER IN DEBATE AT THE UNIVERSITY OF TORONTO

A Customized Graphical Checklist for Efficient Ambulance Inventory

CO-PRESENTER IN "PRAXIS II SHOWCASE" AT THE UNIVERSITY OF TORONTO

Montréal, QC, Canada

May 2019

Cambridge, MA, USA

Apr. 2019

Mississauga, ON, Canada

Jan. 2017

Toronto, ON, Canada

Dec. 2015

Toronto, ON, Canada

Apr. 2014

Teaching

Massachusetts Institute of Technology

TEACHING ASSISTANT

- Fall 2020: 16.485 Visual Navigation for Autonomous Vehicles (Prof. Luca Carlone)

The Carpentries

CERTIFIED SOFTWARE CARPENTRIES INSTRUCTOR

- Oct. 2020: Helper for WHOI Data Carpentry Workshop
- July 2020: Co-Instructor for WHOI Python Workshop Series
- Oct. 2019: Helper for WHOI Software Carpentry Workshop

Worldwide

Aug. 2020 - Dec. 2020

Worldwide

Oct. 2020 - Present

Professional Service

MIT-WHOI Joint Program

ELECTED AT-LARGE REPRESENTATIVE

- Co-hosted monthly online student events to maintain social cohesion during worldwide pandemic
- Developed and published the MIT-WHOI Joint Program online photoboard

Zebra Technologies Inc.

EDITOR, EMC INNOVATION NEWSLETTER

- Edited bi-monthly department newsletter and distributed it to over 1700 engineers
- Commissioned, reviewed, and published articles about recent trends and innovations in electronics, robotics, etc.

Graduate Student Member

IEEE (INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS)

Woods Hole, MA, USA

Oct. 2020 - Present

Mississauga, ON, Canada

May 2016 - Aug. 2017

Worldwide

Nov. 2013 - Present

Academic Service

- Reviewed Journal Submissions for:
 - Robotics and Automation Letters (RA-L)
 - Journal of Aerospace Information Systems (JAIS)
- Reviewed Conference Submissions for:
 - International Conference on Machine Learning (ICML)
 - International Conference on Robotics and Automation (ICRA)
 - International Conference on Intelligent Robots and Systems (IROS)

Personal Interests

Corpus Christi Jazz Horns and Concert Band

SAXOPHONIST

- Performed in the Atlantic Music Festival (2013), Toronto Music Festival (2012)

Burlington, ON, Canada

Sept. 2009 - June 2013